Tool-Lab: Tracing Information Acquisition and Optimal Decision-Making in Large Language Models

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Abstract

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1 Introduction

2 Related Work

3 The Tool-Lab Methodology

4 Experiment 1: Political Science

We instantiate the first Tool-Lab task as a dynamic presidential campaign adapted from Lau & Redlawsk (2006). The original study asked human voters to navigate a mock election in which information arrived as a transient stream rather than as a fixed table: only a few candidate-attribute labels were visible at any moment, opening one item revealed its contents, and the rest of the stream continued to move. This design is useful for Tool-Lab because it turns information acquisition itself into the object of study. The experiment does not ask only which candidate a model selects; it asks which pieces of evidence the model decides are worth spending its limited access opportunities on.

Our environment preserves the core structure of the political-science task. Each episode begins with eight unfamiliar presidential candidates, four Democrats and four Republicans, and proceeds through a primary stage followed by a general-election stage. Candidates are described through the same broad classes of cues emphasized by Lau & Redlawsk (2006): issue positions, party and ideological labels, personal background, personality cues, campaign strategy, endorsements, and polls. The model is assigned a party registration before the primary and can inspect only one currently available item at a time. Inspecting an item is implemented as a tool call. While the model reads that item, new items continue to arrive and previously visible items can disappear, creating an explicit opportunity cost for search. As in the original dynamic board, some cue types are intentionally easier to encounter than others, allowing salient signals such as party and poll standing to compete with more effortful issue-level evidence.

We log the full sequence of tool calls and evaluate the trace using three families of measures from the Mouselab literature. Depth of search captures how much information the model acquires before voting, measured from the number of unique items and distinct attributes inspected. Comparability of search captures how evenly the model examines the same attributes across the candidates in its current choice set. Sequence of search captures whether the model searches candidate-by-candidate or attribute-by-attribute, which allows us to distinguish more systematic compensatory search from shallower or confirmatory heuristics. These process measures are computed separately for the primary and general-election stages so that we can test whether narrowing the field changes the model's search strategy.

Decision quality is evaluated separately from search behavior. Because the campaign is synthetic, each candidate's attributes and the voter profile presented to the model are fully specified in advance, which makes it possible to define a normative correct vote as the candidate that best matches the fully informed preference profile. The political-science experiment therefore yields two outputs for each model: a final vote

Table 1: Summary of Mouselab research across disciplines.

Subject Field	**Article**	**Context**	**Measures**	**Findings**
Accounting & Corporate Finance	Hodge et al. (2004)	Nonprofessional investors evaluated firms using searchable XBRL vs. PDF to allocate a $10k investment.	Information acquisition and integration, investment decisions, and firm performance judgments.	XBRL technology improved nonprofessionals' acquisition and integration of disclosed information.
Consumer Behavior	Payne et al. (1988)	Simulated choice heuristics and human process-tracing using Mouselab evaluating risky multi-attribute options under time pressure.	Information acquisition (amount, speed, sequence, filtration) and relative choice accuracy.	Under high time pressure, users adapted by using attribute-based heuristics, processing faster, and filtering info, maintaining accuracy.
Emotion & Moral Psychology	Luce et al. (1997)	Participants made multi-attribute choices under varying negative emotion conditions (e.g., child charity donation, job choice).	Extent and pattern of processing (time, sequence, alternative vs. attribute search), and self-reported emotion.	Higher negative emotion led to more extensive processing and a shift towards attribute-based, avoiding difficult trade-offs.
Game Theory & Behavioral Economics	Johnson et al. (2002)	Three-round sequential bargaining games with a shrinking pie to track information acquisition and backward induction.	Bargaining behavior (offers, rejection rates) and information search patterns (payoff box lookaheads).	Most subjects exhibited limited lookahead, failing backward induction; instruction improved both lookahead and normative offers.
Healthcare Economics & Policy	Johnson et al. (2013) Dellaert et al. (2024)	Consumers chose health insurance plans with/without aids (calculators, smart defaults); impacts of choice architecture.	Choice accuracy (cost-effectiveness), financial errors, weight on price components, and decision time.	Without decision aids, consumers made large financial errors by overweighting deductibles; smart defaults and calculators improved choices.
Political Science	Lau & Redlawsk (2006)	Voters gathered information and cast votes in a dynamic mock presidential election simulation with scrolling info.	Information search depth, comparability, sequence (candidate/attribute), and matching to decision strategies.	Voters vary in decision strategies (e.g., rational vs. heuristic); dynamic info tracking reveals heuristic use in complex races.
Social Psychology & Intergroup Bias	Freeman et al. (2011)	Categorization of racially ambiguous faces dressed in high-status (business) vs. low-status (janitor) attire.	Categorization choice (White/Black) and real-time computer mouse hand trajectories (Area Under Curve).	High-status attire increased White categorizations, low-status increased Black; trajectories showed implicit attraction to unselected categories.

and a process trace. Comparing them lets us ask whether models that reach the correct choice do so by broad, comparative search or by brittle heuristic shortcuts, and whether harder campaign conditions, such as larger primaries or less ideologically distinctive finalists, push models toward the same kinds of bounded strategies observed in human voters.

4.1 Design

At the level of experimental control, the political-science task is a synthetic two-stage election with known ground truth. Each episode contains four Democratic and four Republican candidates. Each candidate is instantiated from a common inventory of forty-six candidate-specific attributes spanning issue positions, ideology, background, personality, and campaign strategy, together with campaign-level cues such as endorsements and polls (Lau & Redlawsk, 2006). A voter profile is specified in advance and shown to the model at the start of the episode, but the fully informed ranking that it induces over candidates is not. This separation lets us score whether a model votes correctly without confounding the benchmark with the model's own verbal rationale.

The model interacts with the campaign through a constrained inspection interface. During the primary stage, it is assigned a party registration and can inspect only one currently available item at a time before deciding whether to continue searching or cast its vote for an in-party candidate. The general-election stage then presents the two nominees under the same interface. To preserve the logic of the dynamic information board, cue availability is transient: the visible window changes over time, some cue types appear more frequently than others, and inspecting one item can cause other items to be missed. Search therefore carries an explicit opportunity cost, rather than simply revealing a fixed database in arbitrary order.

The main manipulations target decision difficulty rather than surface form. Following the original paradigm, we vary the effective size of the primary choice set and the ideological distinctiveness of the general-election matchup, producing easier and harder campaign conditions without changing the tool interface. All models are evaluated under the same task specification and are compared on both outcome and process variables: their primary and general-election votes, correctness relative to the fully informed benchmark, depth of search, comparability of search, and the sequence of inspection actions. The resulting design makes it possible to distinguish models that arrive at the same electoral choice through systematic comparative search from models that reach that choice through shallow or confirmatory heuristics.

4.2 Procedure

Each evaluation run is a single episode consisting of a briefing, a primary campaign, and a general-election campaign. At the start of the episode, the model receives the voter vignette, its party registration, and the rules of the inspection interface. The vignette defines the goal of choosing the candidate who best matches the assigned voter, but the fully informed ranking implied by all hidden attributes is not revealed. From that point forward, the model can interact only through the Tool-Lab actions: it may inspect one currently visible item or stop searching and cast a vote.

During the primary stage, cue labels arrive in a rolling window and compete for inspection. Opening a cue reveals one candidate- or campaign-level fact, but because the stream continues to advance, other visible items can be missed while the model is reading. The primary stage ends when the model votes for an in-party candidate or when the stage budget is exhausted. The environment then announces the two nominees and resets the stream for the general election, where the model faces the same inspection-and-vote loop over the finalists. Any written justification is collected only after the vote is recorded, so the process trace reflects pre-decision search rather than post hoc explanation.

We repeat this procedure across multiple independent episodes for every model-condition cell, covering both the easier and harder campaign conditions described above. Candidate instantiations, cue order, and moment-to-moment availability are randomized within the design constraints so that models cannot solve the task by memorizing a fixed presentation order. Every action is logged with its stage, cue type, candidate target, and whether the information was new or revisited. These logs are converted into the Mouselab measures of search depth, comparability, and sequence for the primary and general election separately, and

the final votes are scored against the fully informed benchmark. The resulting data pair outcome quality with information-acquisition behavior for every run.

Figure 1 summarizes the episode-level procedure.

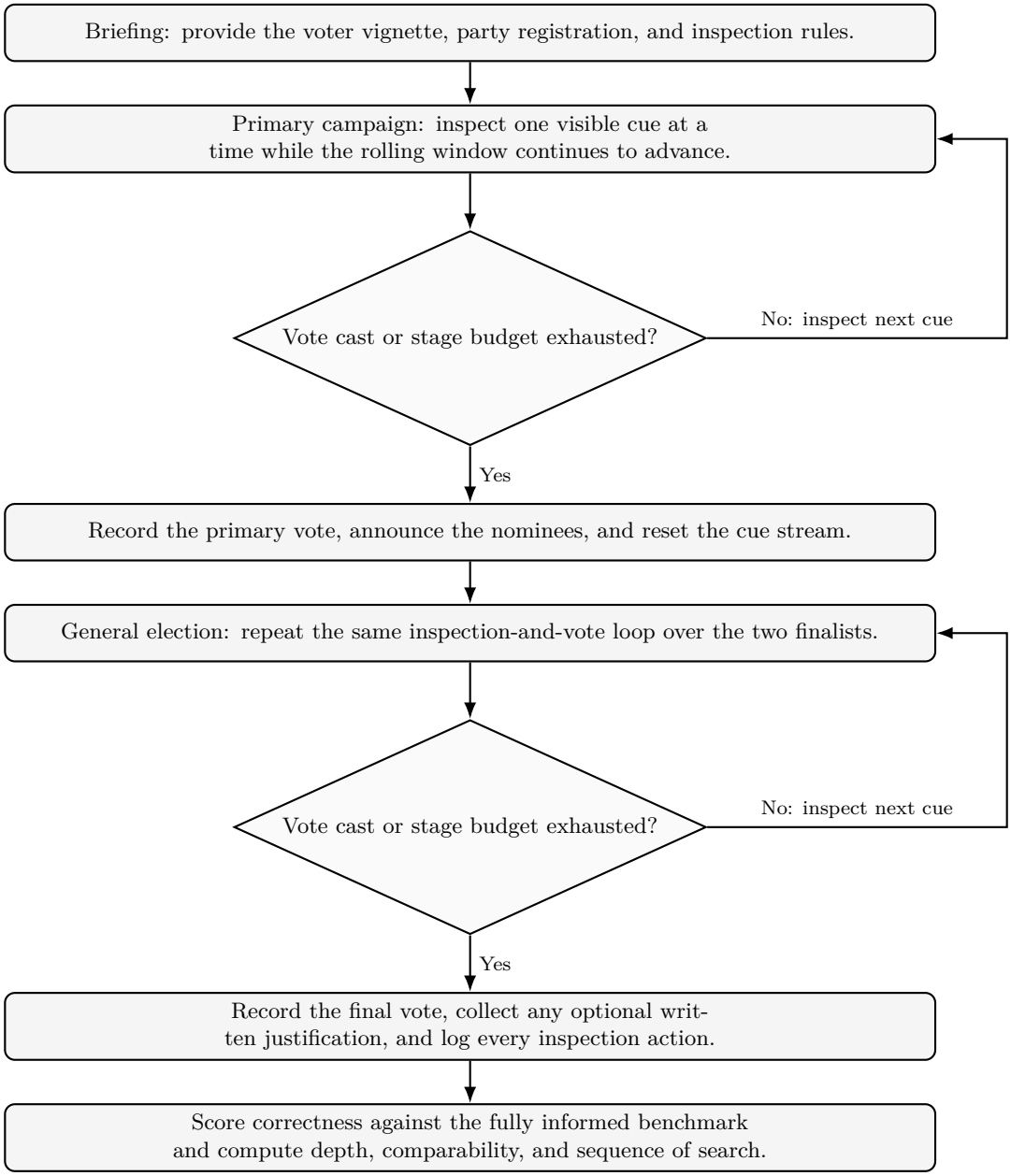

Figure 1: Tool-Lab political-science procedure for one evaluation episode.

4.3 Measures

We evaluate each run at two levels: the quality of the vote that the model ultimately casts and the process by which it acquires information before voting. All measures are computed separately for the primary and general-election stages, with the relevant choice set defined as the candidates who are actually eligible at that stage. In the primary, this means that process measures are restricted to in-party candidates; inspections

of out-party material are logged but do not contribute to the primary search metrics. Repeated openings of the same cue are retained in the raw trace, but they count as revisits rather than new information.

Depth of search measures how much information the model gathers before stopping. Following Lau & Redlawsk (2006), we use three complementary indicators so that the metric is not mechanically driven by the number of candidates in the current choice set: (i) the number of distinct attributes inspected at least once, (ii) the total number of unique cue items inspected for candidates in the current choice set, and (iii) the average number of unique cue items inspected per eligible candidate. For analysis, we report these components directly and combine their within-stage standardized values into a composite depth score. Higher values indicate broader and more persistent search prior to voting.

Comparability of search measures whether the model compares like with like across candidates rather than sampling idiosyncratic facts from each one. We operationalize comparability at the attribute level. An attribute counts as *considered* if the model inspects it for at least one candidate in the current choice set, and it counts as *comparable* if the model inspects that same attribute for every candidate in the set. The comparability score is the proportion of considered attributes that become comparable by this definition. A high score therefore indicates that the model is constructing cross-candidate comparisons on common dimensions, whereas a low score indicates patchier, less comparable search.

Sequence of search measures how the model moves from one inspection to the next. Each adjacent pair of inspections is classified as candidate-based if the model stays with the same candidate and switches attributes, attribute-based if it keeps the same attribute and switches candidates, cross-switch if both candidate and attribute change, or revisit if the identical cue is reopened. Because our interface is dynamic and not every transition is always possible, we normalize candidate-based and attribute-based search by the set of feasible transitions at each step rather than by raw counts alone. This yields a stage-specific candidate-based rate and attribute-based rate, together with a signed orientation score defined as the difference between the two. Positive values indicate a more candidate-based trace and negative values a more attribute-based trace.

Outcome quality is evaluated independently of the process trace. For each stage, the environment defines a fully informed utility ranking over the eligible candidates from the assigned voter profile and the hidden cue values. A vote is scored as correct if the model selects the candidate with the highest fully informed utility; ties are treated as jointly correct. We report correctness as a binary outcome at the primary and general-election stages and retain the full trace so that accurate choices reached by systematic comparative search can be distinguished from equally accurate choices reached by narrow heuristic search. This separation between process and outcome is central to Tool-Lab: it lets us evaluate not only whether a model reaches the right answer, but how it used its limited inspection opportunities to get there.

5 Experiment 2: Consumer Behavior

6 Experiment 3: Finance

7 Discussion and Insights

8 Conclusion and Future Work

References

Benedict GC Dellaert, Eric J Johnson, Shannon Duncan, and Tom Baker. Choice architecture for healthier insurance decisions: Ordering and partitioning together can improve consumer choice. *Journal of Marketing*, 88(1):15–30, 2024.

Jonathan B Freeman, Andrew M Penner, Aliya Saperstein, Matthias Scheutz, and Nalini Ambady. Looking the part: Social status cues shape race perception. *PloS one*, 6(9):e25107, 2011.

Frank D Hodge, Jane Jollineau Kennedy, and Laureen A Maines. Does search-facilitating technology improve the transparency of financial reporting? *The Accounting Review*, 79(3):687–703, 2004.

Eric J Johnson, Colin Camerer, Sankar Sen, and Talia Rymon. Detecting failures of backward induction: Monitoring information search in sequential bargaining. *Journal of economic theory*, 104(1):16–47, 2002.

Eric J Johnson, Ran Hassin, Tom Baker, Allison T Bajger, and Galen Treuer. Can consumers make affordable care affordable? the value of choice architecture. *PloS one*, 8(12):e81521, 2013.

Richard R Lau and David P Redlawsk. *How voters decide: Information processing in election campaigns*. Cambridge University Press, 2006.

Mary Frances Luce, James R Bettman, and John W Payne. Choice processing in emotionally difficult decisions. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 23(2):384, 1997.

John W Payne, James R Bettman, and Eric J Johnson. Adaptive strategy selection in decision making. *Journal of experimental psychology: Learning, Memory, and Cognition*, 14(3):534, 1988.

A Appendix